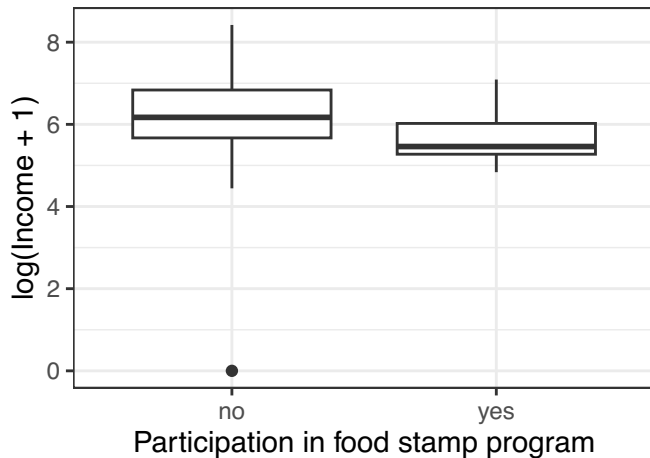


Outliers and multicollinearity

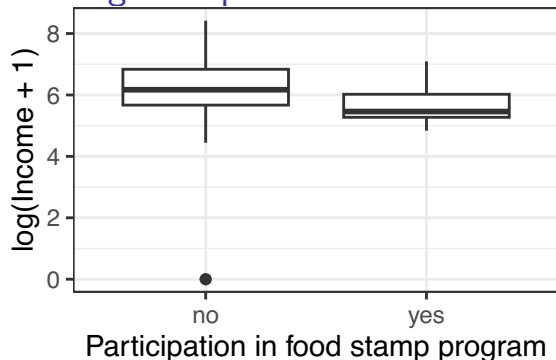
Ciaran Evans

Motivating example



Question: Why are we concerned about outliers when we fit a regression model?

Motivating example



	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.061	1.663	1.239	0.215
log(income + 1)	-0.617	0.278	-2.224	0.026

Question: How would the estimated coefficients change if the potential outlier were removed?

Cook's distance

including potential outlier

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.061	1.663	1.239	0.215
log(income + 1)	-0.617	0.278	-2.224	0.026

excluding potential outlier

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.961	2.276	2.619	0.009
log(income + 1)	-1.274	0.391	-3.257	0.001

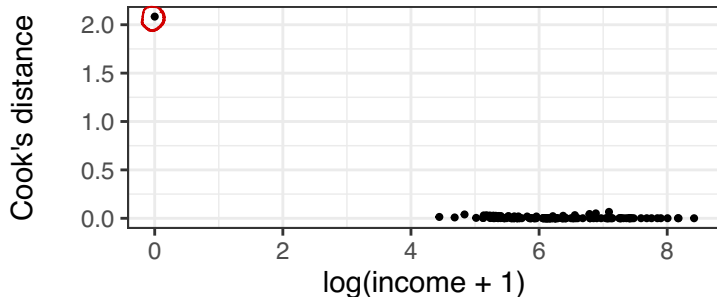
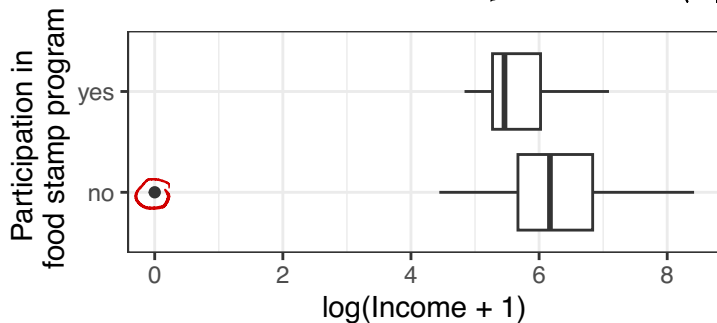
Cook's distance:

- calculated separately for each observation
- measures how much coefficients change if that observation is removed

Cook's distance

General guideline:
distance > 1

concerned if Cook's
(some people use 0.5
instead)



Handling outliers

Should I just remove any unusual observations in the data?

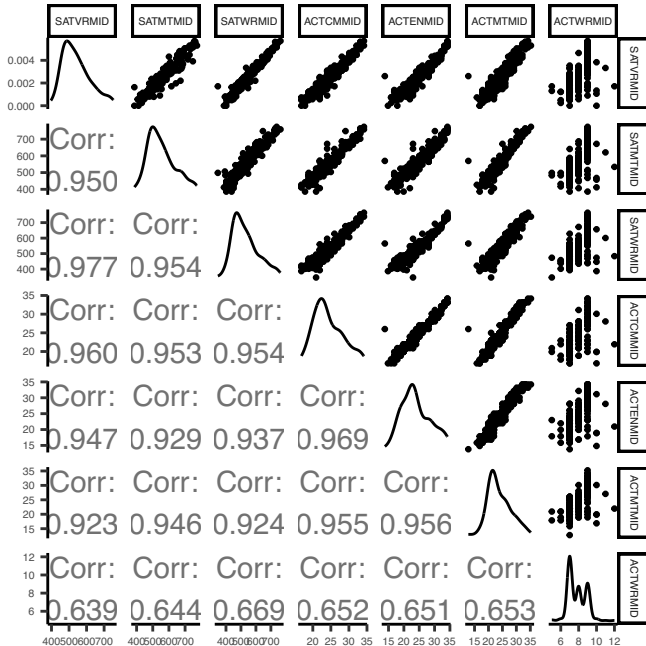
- Always look @ original data!
- Try and identify any errors or missing values
Can safely any clear errors (e.g. weight < 0)
- Can also remove observations from a different population (e.g. a cat in a study on dogs)
- Otherwise: note the unusual observation, report results ($\hat{\beta}$, p-values, CIs, etc.) both with and without observation

College scorecard data

The College Scorecard, compiled by the US Department of Education, is meant to be a tool for college applicants to compare different institutions. Variables include:

- ▶ CONTROL: whether the school is public or private
- ▶ SATVRMID: midpoint of SAT critical reading scores of students attending the school
- ▶ SATMTMID: midpoint of SAT math scores
- ▶ SATWRMID: midpoint of SAT writing scores
- ▶ ACTCMMID: midpoint of the ACT cumulative scores
- ▶ ACTENMID: midpoint of ACT English scores
- ▶ ACTMTMID: midpoint of ACT math scores
- ▶ ACTWRMID: midpoint of ACT writing scores
- ▶ UGDS: number of undergraduate students at the school
- ▶ NPT4: average cost to attend the school
- ▶ RPY_3YR_70: whether the proportion of students actively repaying their loans 3 years after graduation is $> 70\%$ (1 = yes, 0 = no)

College scorecard data ← pairs plot



← high correlations!

⇒ hard to interpret —

now do we change one variable while

holding the others fixed??

Multicollinearity

Multicollinearity: (linear) relationship between 2 or more explanatory variables.

- hard to interpret coefficients
- hard to estimate coefficients
(increased variability)
- Diagnostics:
 - pairs plot
 - correlation between explanatory variables
 - after fitting model:
variance inflation factors (VIFs)

Example: multicollinearity

With all the test score variables:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-11.056	1.930	-5.728	0.000
SATVRMID	-0.005	0.011	-0.485	0.628
SATMTMID	0.001	0.008	0.118	0.906
SATWRMID	0.013	0.010	1.209	0.227
ACTCMMID	0.238	0.188	1.266	0.205
ACTENMID	-0.334	0.153	-2.183	0.029
ACTMTMID	0.370	0.154	2.397	0.017
ACTWRMID	0.036	0.203	0.179	0.858

With just one test score variable:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-9.728	1.350	-7.206	0
SATVRMID	0.018	0.003	7.151	0

Variance inflation factors

$$\text{VIF}(\hat{\beta}_j) = \frac{\text{var}(\hat{\beta}_j \text{ if all variables included in model})}{\text{var}(\hat{\beta}_j \text{ if } X_j \text{ is only explanatory variable})}$$

$$= \frac{1}{1 - R_j^2}$$

$\beta_0 + \beta_j X_j$

R^2 for regression of X_j on all other explanatory variables

as $R_j^2 \uparrow$ (better able to predict X_j from the other explanatory variables)

$\text{VIF} \uparrow$

$$1 \leq \text{VIF} \leq \infty$$

common guidelines: concerned if $\text{VIF} \geq 5$ ($R_j^2 = 0.8$)
or $\text{VIF} \geq 10$ ($R_j^2 = 0.9$)

Handling multicollinearity

```
m1 <- glm(RPY_3YR_70 ~ .,  
          family = binomial, data = scorecard)
```

```
car::vif(m1)
```

```
##    CONTROL  SATVRMID  SATMTMID  SATWRMID  ACTCMMID  ACTEN  
##  3.539098 19.586976 12.138599 18.767795 13.158163 12.693  
##      UGDS  PCTFLOAN      NPT4  
##  2.511964  2.124875  2.438207
```

What should we do when there is high multicollinearity?

Summary

- ▶ Outliers:
 - ▶ Change our coefficient estimates
 - ▶ Identify with EDA before model fitting
 - ▶ Also check with Cook's distance after model fitting
 - ▶ Clear errors, or points from a different population of interest, can be removed. Otherwise, report results with and without outliers
- ▶ Multicollinearity:
 - ▶ Makes it hard to interpret meaning of a coefficient (how do we hold other variables constant??)
 - ▶ Increases variability of estimated coefficients
 - ▶ Identify with pairs plots, correlation matrices before model fitting
 - ▶ Also check with VIFs after model fitting
 - ▶ Options: use a subset of the explanatory variables; combine several explanatory variables (average, PCA, etc.); ignore (if only predicting)

Class activity

- ▶ Work on the class activity in R (linked on Canvas and the course website)
- ▶ You are welcome to work in groups
- ▶ At the end of class, submit HTML file on Canvas (one per group, listing all group member names)